

Holographic Memory System Technical Report

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Conversational memory remains difficult because evidence needed for an answer may be scattered across sessions and bear little surface resemblance to the query. Existing systems often reduce recall to a fixed similarity search process, with no explicit control over evidence coverage or organization. To address this problem, we propose the Holographic Memory System (HMS), which separates retention from recall and treats recall as evidence construction guided by each query. HMS compiles each request into a recall program that plans evidence expansion, verifies coverage, and organizes the retrieved memories before generation. Under a unified backbone and evaluation protocol, HMS establishes a new state of the art in overall accuracy on LongMemEval and LoCoMo. These results highlight evidence construction during recall as a key design choice for conversational memory over long interactions.

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Code: <https://github.com/Shadow-Weave/HMS>



1 Introduction

Long-horizon conversational agents must answer questions whose evidence is scattered across many sessions and time points, and that evidence often looks nothing like the question that needs it. Asked “how many events did I attend last month?”, an agent must resolve “last month” relative to the query date rather than to any stored record, gather event records that share almost no tokens with the question, and then count them without double counting near-duplicates. The information is rarely absent. However, it is difficult to access the memory in a form the model can actually reason over under a fixed context budget.

Recent work frames this problem as an agent-memory life cycle in which traces are ingested, written into persistent records, maintained over time, and retrieved for later use (Zhang et al., 2024b; Chhikara et al., 2025; Xu et al., 2025). The framing has been productive, but it entangles the stages. When a system reports higher accuracy, it is rarely clear whether the gain comes from better extraction, better storage, or better retrieval, and systems studies show that write path, substrate, and retrieval flow can differ by orders of magnitude even when downstream accuracy looks similar (Omri et al., 2026). Retrieval in particular is usually treated as a solved commodity, a standard hybrid search over whatever was stored, and is rarely isolated and optimized on its own terms. The field therefore lacks a clean answer to a basic question: holding the stored memory fixed, how much downstream accuracy is left by retrieval alone?

This paper answers that question directly. We decouple retention from recall: a standard retention stage builds the memory bank once, and all optimization happens on the recall side, evaluated against that fixed bank so that every accuracy change is attributable to retrieval behavior rather than to stochastic extraction or differing memory coverage. We stress that this is an evaluation discipline, not a system limitation. In deployment, HMS pairs the retention stage with the retrieval stack introduced below and operates as a complete memory system, and the same discipline is what allows us to report an explicit extraction ceiling alongside every result. The name reflects this design: the evidence a query needs is distributed across the whole bank rather than localized in query-similar records, and each query reconstructs its own organized view of that same fixed content.

Under this lens, standard hybrid retrieval leaves four recurring failures on conversational benchmarks such as LongMemEval and LoCoMo (Wu et al., 2025a; Maharana et al., 2024). First, answers often sit in records that are relevant but dissimilar to the query, such as a neighboring unit from the same session or a record sharing an entity but no surface tokens, and top- k similarity systematically misses them. Second, running every retrieval channel on every query is wasteful and sometimes harmful, since a confidently wrong temporal or graph channel injects noise into queries that

never needed it. Third, when recall genuinely falls short, nothing checks sufficiency before generation, so the burden of recovering from missing evidence lands silently on the reader. Fourth, even when everything needed is retrieved, it arrives as a score-ordered pile in which duplicates recur, planned and completed events interleave, and occurrence time mixes with mention time, and over such a pile counting, ordering, and comparison routinely fail.

As illustrated in Figure 1, conventional hybrid retrieval may spend its fixed context budget on topically similar records, whereas HMS explicitly constructs, verifies, and organizes the evidence required by the query.

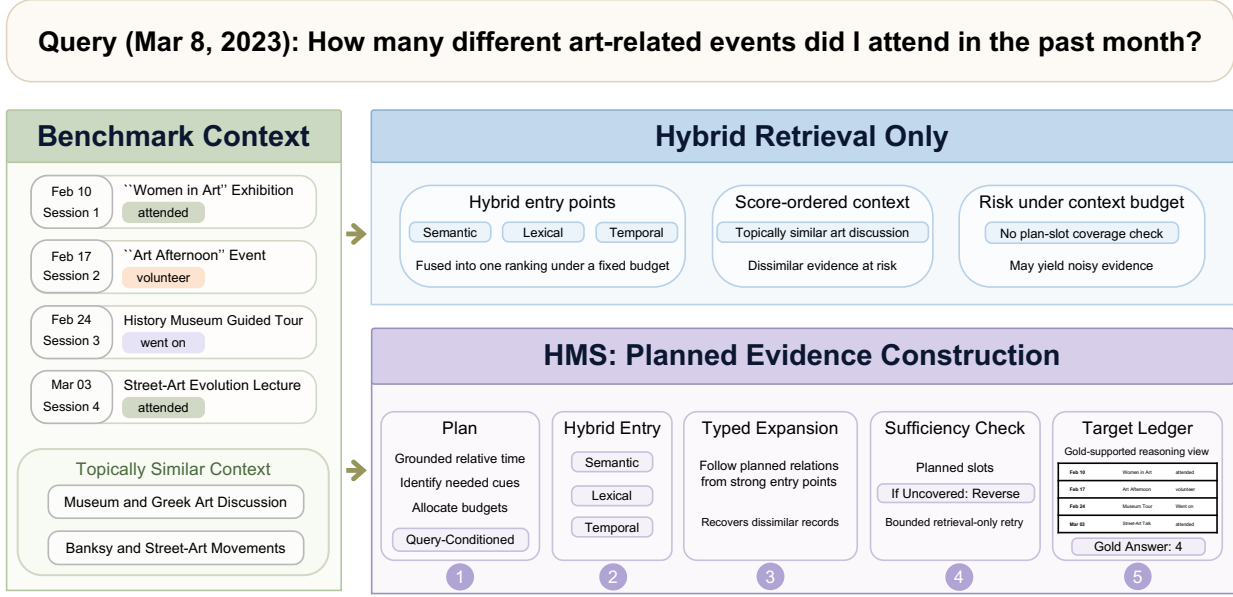


Figure 1 Motivation and overview of the Holographic Memory System. Conventional hybrid retrieval combines multiple retrieval channels into a single ranking under a fixed context budget, which may prioritize superficially similar memories while omitting evidence required for the answer. HMS instead formulates recall as query-conditioned evidence construction. It identifies the required evidence, expands retrieval through typed relations, verifies evidence coverage, and organizes the selected memories into a structured representation for downstream reasoning.

HMS addresses these failures through one principle: move reasoning-critical decisions earlier, into evidence construction. The design echoes the staged character of human recall, which forms a retrieval plan from the question and its cues before searching, monitors whether enough has been recalled, and reconstructs an organized account before reporting (Tulving and Thomson, 1973; Nelson and Narens, 1990). Concretely, a recall planner grounds underspecified queries and compiles each into an executable operator program. A library of six typed evidence operators, covering session locality, entity bridges, temporal chains, knowledge updates, contradictions, and multi-hop structure, recovers relevant-but-dissimilar evidence by propagating an entry point’s strength rather than the query’s similarity. A sufficiency verifier checks the recalled evidence against the plan and triggers bounded retries that stay on the retrieval side. A gated organization layer finally reshapes the verified evidence into a deduplicated, time-explicit ledger paired with raw source snippets. The planner runs deterministically and every stage operates over the fixed bank, so the whole pipeline remains attributable retrieval behavior.

We make the following contributions:

- **A query-time evidence-construction framework for conversational memory.** HMS couples a recall planner with a library of six typed evidence operators under a shared score-propagation interface, so that channel selection, budget allocation, and relation type become per-query decisions compiled into an executable program rather than fixed architecture (Sec. 3.3 and Sec. 3.4).
- **A retrieval-side loop for sufficiency and structure.** A verifier diagnoses coverage gaps against the plan and triggers bounded retries, and a gated organization layer renders the verified evidence as a structured, source-grounded ledger. Together they form a self-correcting retrieval stage that we show is more stable than answer-side verification (Sec. 3.5).

- **A controlled methodology that decouples retention from recall.** By holding memory content fixed and constraining every component to a retrieval-only action space, the methodology makes gains attributable, reports them against a stated extraction ceiling, and lets HMS layer onto an existing retention pipeline without modifying it (Sec. 3.2).
- **State-of-the-art retrieval results with component-level evidence.** On LongMemEval and LoCoMo, HMS improves overall accuracy by 4.0% and 1.4% over a strong hybrid baseline and outperforms the strongest prior memory system under matched conditions, and per-component, per-category ablations show each component improving its matched question type at a reported retrieval cost (Sec. 4).

2 Related Work

2.1 From Retrieval to Memory

Early external-memory work separates knowledge access from model weights. Dense retrieval, reader-generator architectures, and retrieval-augmented generation let a model condition on evidence selected from a large non-parametric store (Karpukhin et al., 2020; Guu et al., 2020; Lewis et al., 2020; Izacard and Grave, 2021). Later systems move retrieval closer to generation through nearest-neighbor datastores, retrieved token chunks, or in-context grounding documents (Khandelwal et al., 2020; Borgeaud et al., 2022; Ram et al., 2023). This line gives modern memory systems their basic read primitive: select compact external evidence instead of relying only on parametric knowledge.

The same limitation appears when retrieval is replaced with longer context. Memorizing Transformers and LongMem extend access to past tokens or cached histories, while long-context benchmarks show that simply placing more material in the prompt still leaves failures under positional bias, multi-hop reasoning, and growing sequence length (Wu et al., 2022; Wang et al., 2023b; Liu et al., 2024; Bai et al., 2023; Zhang et al., 2024a; Hsieh et al., 2024). Retrieval and long context therefore answer the access question, but not the state question. They say how a model can read old information. But they do not decide what should be written, revised, forgotten, or compacted into durable state.

2.2 Memory as an Agent Lifecycle

LLM-agent memory shifts the problem from evidence access to state maintenance. Recent surveys frame agent memory as a write-manage-read loop: the agent ingests interaction traces, transforms them into persistent records, maintains those records over time, and retrieves them for later action (Zhang et al., 2024b; Du, 2026; Yang et al., 2025b). Existing systems instantiate different points in this design space. Some store observations, summaries, reflections, user facts, or learned lessons for planning and personalization (Park et al., 2023; Shinn et al., 2023; Zhong et al., 2023; Zhao et al., 2023). Others preserve reusable procedures or manage virtual context across memory tiers (Wang et al., 2023a; Packer et al., 2023; Sumers et al., 2024). The important pattern is that memory becomes part of the agent’s control state, not a passive cache.

Recent method papers sharpen this lifecycle into several design axes. Flat retrieval memories emphasize efficient storage and recall. Structured memories add graphs, linked notes, multi-granularity records, or typed stores. Policy-managed memories decide when to write, merge, evict, or route records (Chhikara et al., 2025; Xu et al., 2025; Li et al., 2025; Gao and Zhang, 2024; Lu et al., 2026; Hu et al., 2026a; Xu et al., 2026a; Chen et al., 2024; Yang et al., 2025a). Systems studies add a complementary view: write-path construction, storage substrate, retrieval flow, mutability, latency, and energy can vary by orders of magnitude even when downstream accuracy looks similar (Omri et al., 2026). Our work follows this lifecycle view. The relevant object is a governed memory layer that must stay useful as interactions accumulate and context budgets remain fixed.

2.3 Evaluating Durable State

Evaluation has begun to separate long-context ability from durable memory. Long-context benchmarks test whether a model can use material already provided in the prompt (Bai et al., 2023; Zhang et al., 2024a; Hsieh et al., 2024). Memory benchmarks impose a stricter condition: useful information must survive across sessions, updates, and task boundaries. LoCoMo and LongMemEval test multi-session conversational memory, temporal reasoning, knowledge updates, and abstention (Maharana et al., 2024; Wu et al., 2025a; Hu et al., 2026b). MemoryAgentBench and MemoryArena extend this toward incremental interaction, selective forgetting, and interdependent agentic tasks (Hu et al., 2025a; He et al.,

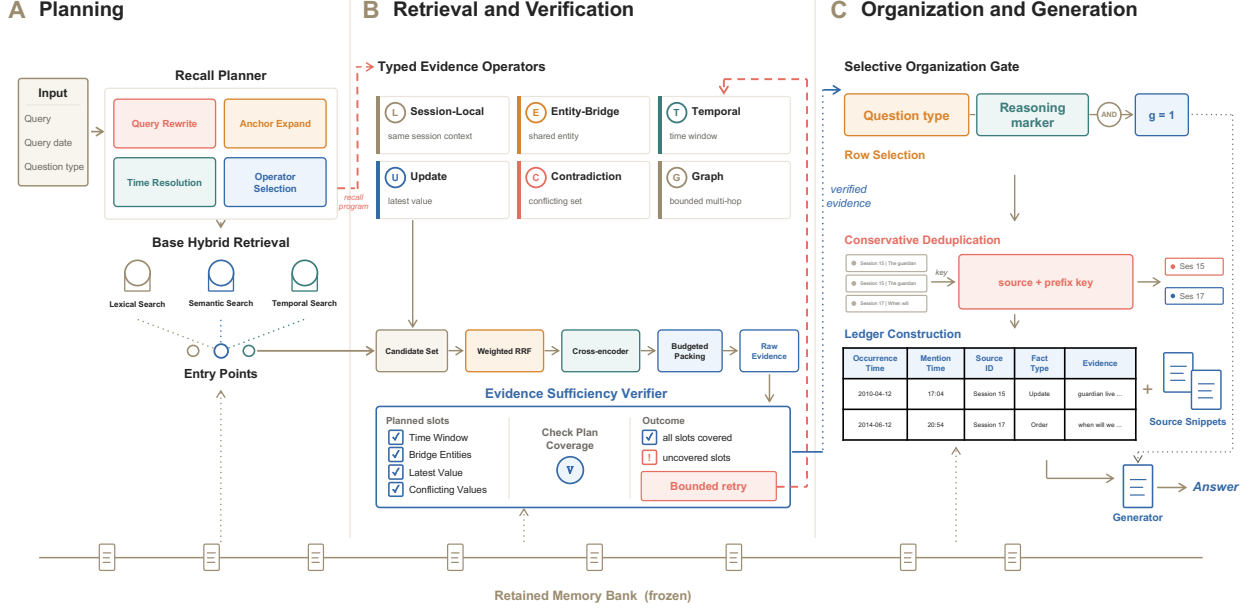


Figure 2 Architecture of the Holographic Memory System. (A) A recall planner translates the input query into a structured retrieval plan over a frozen memory bank. (B) Typed evidence operators retrieve and rank candidate evidence, while a sufficiency verifier iteratively checks whether the required information is covered. (C) Verified evidence is selectively organized into a deduplicated ledger with source grounding, which supports final answer generation.

2026). MemGym makes the evaluation pressure more explicit by isolating the memory module from the reasoning model across tool-use dialogue, coding, search, and web-control settings (Xu et al., 2026b).

This progression changes what a memory method must prove. Full-context baselines ask whether history is useful when it is handed to the model. Retrieval baselines ask whether similar text can be recovered. Durable-memory evaluation asks whether a system preserves task-relevant state, updates it when the world changes, avoids stale evidence, and exposes compact information that changes downstream behavior. We position our method against this stronger contract.

3 Method

3.1 Problem Setup and Pipeline Overview

Figure 2 presents the end-to-end architecture of HMS. The system first compiles the query into a recall program, retrieves and expands candidate evidence through selected operators, verifies coverage against the planned requirements, and selectively organizes the verified evidence before generation. All stages operate over the same fixed memory bank.

Memory bank. HMS operates over a memory bank, the standard substrate of long-horizon conversational memory systems (Latimer et al., 2025; Chhikara et al., 2025). An extraction pipeline reads a conversational corpus $\mathcal{D}$ once and distills it into a set of self-contained memory units $\mathcal{M} = \{m_1, \dots, m_N\}$. Each unit is a short narrative statement about the user paired with the metadata needed to reason about it later,

$$m = (\text{id}, t, v, [\tau_s, \tau_e], \tau_{\text{ment}}, \sigma, \mathcal{E}_m),$$

where t is the narrative text, $v \in \mathbb{R}^d$ its embedding, $[\tau_s, \tau_e]$ the interval during which the described event occurred, τ_{ment} the time at which the user mentioned it, σ the session identifier, and $\mathcal{E}_m$ the set of canonical entities the unit mentions. We write $\text{tokens}(m)$ for the token length of t . The distinction between occurrence time and mention time, and the retention of session and entity structure, are what later stages exploit.

Memory retention. The bank is built by the retain stage of Hindsight (Latimer et al., 2025), which we adopt unchanged and for which we claim no novelty. An LLM reads each session as a whole and distills it into a small number of

stand-alone units, each covering an entire exchange rather than a single utterance, so that participants, their roles, and any cross-turn reasoning survive inside one unit instead of being scattered over many. During this pass, relative time expressions are converted into absolute occurrence intervals and kept separate from the timestamp of the mention itself, a type is assigned to each unit, and the entities it refers to are recorded. Surface mentions of the same referent are then consolidated into canonical entity identifiers using a mix of name similarity, co-occurrence statistics, and closeness in time, and the finished units are wired together at indexing time through shared entities, temporal adjacency, and embedding similarity. The co-occurrence structure traversed by the G operator of Sec. 3.4 is derived from this index. The retain stage can additionally emit explanation-oriented links between units, which HMS does not rely on, for reasons discussed in Sec. 3.4. Everything downstream of retention is ours: HMS takes the retained bank as given and changes only how it is accessed.

Given a query q issued at date d_q with question type y and a token budget B , the system must return a compact evidence set

$$\mathcal{E}_q \subseteq \mathcal{M}, \quad \sum_{m \in \mathcal{E}_q} \text{tokens}(m) \leq B,$$

from which a downstream reader generates the answer. Crucially, a useful $\mathcal{E}_q$ is not merely a set of units individually similar to q . It must be temporally grounded, session consistent, and relationally connected, and it must reach the reader in a form over which counting, ordering, comparison, and state resolution are possible without further guesswork.

Why retrieval is the bottleneck. In real human-agent dialogue scenarios, agents must process long-context and fragmented dialogue histories to respond to user queries, which presents challenges of ensuring effective multi-session information aggregation, precise temporal reasoning, and knowledge disambiguation as the conversation unfolds. Existing memory systems based on hybrid retrieval often fail when confronted with these challenges. However, their failure typically arises not from missing required evidence, but from a mishandled recall process. Although these systems incorporate multiple retrieval channels, they commonly rely on a monolithic similarity metric unrelated to the task and apply all available channels uniformly to every query. This approach yields noisy retrieval results and lacks the structured organization required for complex reasoning. Consequently, the model struggles to accurately aggregate and reason over missing, conflicting, or poorly organized evidence, frequently resulting in omissions or totally incorrect outputs. The bottleneck can be attributed to two main issues in the retrieval phase. First, current approaches fail to exploit the latent information inherent in user questions, specifically the intent underlying the question and the forms of relatedness it entails. Second, the retrieved evidence often lacks sufficiency and internal consistency, and even when present, it is rarely organized in a manner conducive to downstream LLM reasoning. Owing to these deficiencies, existing memory systems remain ill-equipped for the demands of real-world human-agent conversations.

Guiding principle. Our design takes its cue from how human memory handles the same problem. Cognitive research characterizes human recall not as a single associative lookup but as a staged, monitored process: a retrieval plan is formed from the question and its cues before search begins (Tulving and Thomson, 1973), search then proceeds along qualitatively different associative routes such as temporal order, shared participants, and spatial or episodic context (Kahana, 2012), metamemory continuously monitors whether what has been recalled is enough and re-initiates search when it is not (Nelson and Narens, 1990), and the recalled fragments are finally reconstructed into an organized account rather than reported as a raw stream (Bartlett, 1932). Inspired by this staged view, HMS follows one principle: *move reasoning-critical decisions earlier in the pipeline*. Instead of retrieving indiscriminately and repairing at generation time, HMS first plans what evidence a query needs, executes only the typed evidence operators the plan calls for under per-operator budgets, verifies that the recalled evidence covers the plan, and organizes the verified evidence into a reasoning-ready form before the reader ever sees it. This reconstructive view is also what the system’s name records: the required evidence is distributed over session neighborhoods, entity bridges, and temporal chains rather than localized in query-similar records, and the planner, operators, and ledger reconstruct an organized view of it for each query.

Roadmap. Each stage of this principle targets a concrete failure mode and is presented in its own subsection. The recall planner (Sec. 3.3) grounds underspecified queries and compiles them into an executable recall program, so that channel selection and budget allocation become per-query decisions rather than fixed architecture. The typed evidence operators (Sec. 3.4) replace a single notion of relatedness with a library of relation-specific expansions, recovering evidence that is relevant but dissimilar to the query. The post-retrieval stage (Sec. 3.5) then closes two remaining gaps on the retrieval side, using a sufficiency verifier against under-recall and a structured organization layer against disorganized evidence. Sec. 3.6 positions these choices against prior systems, and Sec. 4 evaluates each stage separately.

Pipeline at a glance. Figure 2 shows the full pipeline, which the following subsections instantiate. For each query, the planner maps (q, y, d_q) to a recall program. Entry points are retrieved from three parallel base channels, semantic, lexical, and temporal, a now-standard hybrid backbone that we deliberately do not claim as a contribution (Latimer et al., 2025; Chhikara et al., 2025; Sarmah et al., 2024). The planned typed operators then expand from these entry points under their budgets. All candidate lists are fused with weighted Reciprocal Rank Fusion,

$$\text{RRF}(m) = \sum_{\text{source } j} \frac{\omega_j}{\eta + \text{rank}_j(m)},$$

rescored by a cross-encoder reranker, and greedily packed into $\mathcal{E}_q^{\text{raw}}$ under the budget B . The sufficiency verifier then checks $\mathcal{E}_q^{\text{raw}}$ against the plan and, if slots are uncovered, revises the operator program and re-executes within a bounded number of rounds. Finally, for queries that need it, the organization layer reshapes the verified evidence into a structured ledger with source grounding, and the result is handed to the reader.

3.2 Decoupling Retention from Recall

HMS treats memory acquisition and memory access as two separate optimization problems and commits to improving only the latter. Concretely, the retained bank $\mathcal{M}^* = \text{Retain}(\mathcal{D})$ is built once by the retention stage of Sec. 3.1 and then frozen, and every candidate retriever R_i is evaluated as $\mathcal{E}_q^{(i)} = R_i(q, d_q, y, \mathcal{M}^*, B)$ against the same $\mathcal{M}^*$. We freeze the bank for three reasons. First, *attribution*. Extraction is stochastic and expensive, so if a system changes extraction and retrieval together, an accuracy gain cannot be assigned to either, and re-running extraction per variant silently changes memory coverage between comparisons. Freezing makes any downstream difference attributable to retrieval and evidence organization alone. Second, *comparability*. All ablations in Sec. 4 see an identical set of stored memories, so the per-component ablation is a controlled comparison rather than a re-run of extraction. Also noteworthy, this freezing only applies to our proposed system and its ablations. All baseline methods are evaluated in their original end-to-end settings without any such freezing. Third, *deployment realism*. In practice a memory bank accumulates over months of user interaction and cannot be re-extracted every time the retrieval algorithm changes, so retrieval methods that improve accuracy over a fixed bank are exactly the methods that transfer to deployed systems.

To make the protocol precise we fix the action space available to a retriever. An admissible action may read $\mathcal{M}^*$, score and rank its units, expand from selected units to other units already in $\mathcal{M}^*$, reorganize the selected units, and re-issue any of the above under a revised plan or budget. It may not add, edit, or re-extract units. Every component of HMS, including the planner, the verifier-triggered retries, and the organization layer, is constrained to this action space, which is what keeps the bank effectively frozen even when the pipeline re-plans or reorganizes evidence. Because the planner (Sec. 3.3) may be realized by an LLM, we treat its output as part of retrieval behavior rather than as a second source of extraction noise: unless otherwise stated it runs at temperature 0 with a fixed prompt, so a given (q, d_q, y) maps to a reproducible plan, and we report plan stability across seeds in Sec. 4. The protocol is realized by a single benchmark entry point operating in a retrieval-only mode that skips ingestion, reuses the existing memory bank, and writes to fixed data, log, and result directories, so every variant is a recoverable configuration over one frozen $\mathcal{M}^*$.¹

We state the protocol’s ceiling explicitly rather than leaving it as an unspoken flaw. Because $\mathcal{M}^*$ is fixed, recall can recover only evidence that extraction already captured, so achievable accuracy is upper bounded by the coverage of $\mathcal{M}^*$, and we report this extraction ceiling alongside our results (Sec. 4). Neither the verifier nor the organization layer of Sec. 3.5 can break this ceiling. They can only reallocate retrieval effort toward, or reorganize, evidence that already exists in $\mathcal{M}^*$.

3.3 Recall Planner

As we have analyzed, one critical bottleneck at the retrieval stage lies in the failure to recognize the user’s underlying intent and perform precise temporal reasoning. This shortfall becomes particularly evident when queries require localizing information within a specific time window and aggregating evidence across multiple episodes. For example, a realistic question such as “How many events did I attend last month?” shares few surface tokens with the stored events, and “last month” must be resolved relative to d_q rather than to the mention date of any memory. Resolving these after

¹Implementation: a single benchmark script exposes each variant as a command-line flag, and a retrieval-only switch bypasses ingestion and reuses the persisted memory store. Concrete flags and directory layout are given in the supplementary material.

retrieval cannot recover evidence that was never retrieved. Equally, running every channel on every query wastes budget and lets a confidently wrong channel inject noise into queries that never needed it.

We therefore ground the query before retrieval and let that grounding decide which operators run and how much budget each receives. A lightweight planner maps (q, y, d_q) to a structured recall plan

$$A(q, y, d_q) = \left(q', \mathcal{X}, [t_s, t_e], \{(o, \theta_o, \omega_o, b_o)\}_{o \in \mathcal{O}_q} \right),$$

where q' is a rewritten query, $\mathcal{X}$ expands abstract category terms into concrete retrieval anchors, and $[t_s, t_e] = \text{TemporalResolve}(q, d_q)$ is an optional absolute time window. The core element is the operator program $\{(o, \theta_o, \omega_o, b_o)\}$: a subset $\mathcal{O}_q \subseteq \mathcal{O}$ of the typed evidence operators of Sec. 3.4, each with per-query parameters θ_o , a fusion weight ω_o , and a retrieval budget b_o . The plan is thus an executable specification of what to retrieve, how much of it, and how to combine it.

The plan is conditioned on the question type y , which controls how far expansion reaches. In our configuration the fusion component reduces to a per-type cross-session expansion coefficient $\lambda_y \in [0, 1]$, chosen to increase monotonically with the cross-session demand of the type,

$$\lambda_{\text{single}} < \lambda_{\text{preference}} < \lambda_{\text{update}} < \lambda_{\text{temporal}} < \lambda_{\text{multi}},$$

so single-session queries stay narrow while multi-session and temporal queries receive broad cross-session coverage.² This schedule is the concrete, frozen instantiation of the plan’s fusion weights, and the operator program generalizes it to per-operator weights and budgets.

Two further mechanisms make the plan adaptive rather than merely selective. *Query-adaptive budget allocation* partitions the global token budget across the selected operators,

$$\sum_{o \in \mathcal{O}_q} b_o \leq B, \quad b_o \geq 0,$$

so the plan can concentrate retrieval effort where the query most needs it rather than spending a fixed budget per channel. Intuitively, b_o caps how many candidates operator o may contribute before fusion, whereas ω_o governs how those candidates are weighted once contributed. The two are complementary, since a high weight cannot help evidence an operator was never allowed to retrieve. Confidence-adaptive weighting lets the planner emit a per-operator confidence that becomes the operator’s fusion weight, so unreliable signals are down-weighted rather than silently trusted. For the temporal operator, $\omega_T = \omega_0 c_T$ with $c_T \in [0, 1]$ the confidence that a well-formed window was extracted, and the remaining mass is redistributed over the operators $\mathcal{O}_q$ retains. When no reliable temporal anchor exists the planner omits the temporal operator entirely (equivalently $b_T = 0$), which is strictly stronger than running it at low weight: a confidently wrong window can no longer suppress relevant evidence because it is never retrieved.

3.4 Typed Evidence Operators

Why one traversal rule is not enough. Recent memory systems already expand recall beyond raw similarity by walking a memory graph. Hindsight (Latimer et al., 2025), for instance, wires units together at ingestion time with entity, temporal, semantic, and LLM-asserted causal edges, and recalls over this graph with a single spreading-activation pass in which every edge type feeds the same activation score under fixed per-type multipliers, with the causal edges upweighted. Graph-augmented RAG systems inherit a similar design (Sarmah et al., 2024). This helps, but the traversal itself is query independent: every question walks the same graph under the same rule, the causal edges commit at ingestion to an explanatory reading that a given query may never ask for, and some relations an answer needs are not ingestion-time edges at all. On LongMemEval we find that the situations in which stored units support an answer are heterogeneous, and each demands a different traversal key. (i) *Fragmented context*: the answer lies in a unit adjacent to a high-scoring entry point in the same session, topically related yet query dissimilar. The key is session position, and a generic link either misses the neighbor or buries it among unrelated associations. (ii) *Cross-unit bridging*: the answer requires joining units that share an entity but no surface tokens. The key is entity identity. (iii) *Knowledge updates*: the current answer is the latest of several units about the same fact. The key is mention recency, an ordering that no symmetric association can

²The specific coefficients used in all experiments, together with the marker lexicon of Sec. 3.5 and the packing constants there, are listed in the supplementary material for reproducibility.

express. (iv) *Conflicting evidence*: two or more units about the same fact disagree, and a reader shown only one answers from a partial view. The key is value disagreement, and the correct behavior is to surface the whole conflicting set, not the single best-linked member. The last two keys, mention recency and value disagreement, are query-time computations over groups of units, not static edges, so no ingestion-time graph can encode them. Folding all four patterns into one graph and one propagation rule thus both overstates the mechanism, since nothing explanatory is verified at query time, and prevents asking which relation actually helps which question category.

A shared operator interface. HMS therefore factors evidence expansion into a library of typed operators sharing one interface. Each operator o maps an entry point p to scored, typed neighbors,

$$o : (p, \mathcal{M}^*, \theta_o, b_o) \rightarrow \{(m_j, w_j, r_j)\}_{j=1}^{k_o},$$

where r_j is the relation type the operator asserts, $w_j \in [0, 1]$ the link strength, θ_o the per-query parameters supplied by the plan, and k_o is bounded by the allocated budget b_o . Critically, every operator propagates the entry point’s strength rather than the neighbor’s query similarity,

$$s_o(m_j) = \lambda_o s(p) w_j \beta(r_j) \gamma_o^{d(p, m_j)},$$

where $s(p)$ is the entry point’s fused score, $\beta(r_j)$ a type-specific boost, $\gamma_o \in (0, 1]$ a decay, and $d(p, m_j)$ a hop or position distance. Propagating $s(p)$ instead of $\text{sim}(q, m_j)$ is the shared design invariant, and it is exactly what lets relevant-but-dissimilar evidence survive fusion: a unit that would score near zero against the query inherits credibility from the entry point that vouches for it.

The operator library. The library comprises six typed operators,

$$\mathcal{O} = \{L, E, T, U, C, G\},$$

each instantiating the interface with a different traversal key.

- **Session-local (L)** targets fragmented context. It adds same-session neighbors of p , with $d(\cdot) = |\text{pos}(p) - \text{pos}(m')|$ the within-session position distance under a Gaussian proximity kernel, so context is recovered by position rather than similarity. To guarantee that a strong entry point never loses its immediate context, the score of directly adjacent neighbors is floored at a fixed fraction of $s(p)$.
- **Entity-bridge (E)** targets cross-unit bridging. It propagates from p to units sharing a canonical entity in $\mathcal{E}_p$, where entity canonicalization is performed once at indexing time so that surface variants of the same referent resolve to one identifier. The parameters θ_o carry the bridge entities named by the plan, so the operator follows the entities the question is actually about rather than every entity p happens to mention.
- **Temporal (T)** supports before and after chains. It restricts propagation to neighbors whose occurrence intervals fall inside the plan’s resolved window $[t_s, t_e]$ and orders them by occurrence time, so the expansion respects the timeline the question refers to instead of the order in which things were said.
- **Update (U)** targets knowledge updates. It groups units that describe the same fact, orders the group by mention time τ_{ment} , and boosts the most recent member, so knowledge-update questions resolve to the current value rather than to whichever historical value is most similar to the query.
- **Contradiction (C)** targets conflicting evidence. It detects units about the same fact whose values disagree and surfaces the conflicting set jointly, each member annotated with its mention time and source. It differs from U in both intent and output: U assumes a single time-ordered truth and collapses the group to its most recent member, whereas C preserves the disagreement so the reader can adjudicate it. To stay within the retrieval-only action space it only retrieves and marks conflicting evidence. It never deletes, edits, or arbitrates between units.
- **Graph (G)** targets multi-hop questions whose links are neither purely temporal nor single-entity. It operates on the co-occurrence structure that the retention index of Sec. 3.1 already provides: a graph whose nodes are memory units and whose edges connect units that share a canonical entity or a session, weighted by co-occurrence frequency. G performs bounded-hop traversal over this precomputed structure from the entry points, with the decay $\gamma_o^{d(p, m_j)}$ applied per hop, so compositional links such as entity-to-session-to-entity chains become reachable at controlled cost.

Because every operator plugs into the same interface and is selected and budgeted by the plan, “which relation type helps” becomes a controlled experimental dimension (Sec. 4) rather than an architectural commitment. The default program for a query with no special structure is $\{L\}$ over the semantic entry points, which keeps simple lookups cheap.

3.5 Post-Retrieval Sufficiency and Organization

Recall, however well planned, leaves two symmetric risks on the table. The recalled evidence may be *too little*: the plan may target an entity the first pass missed, or resolve a window slightly too narrow, and passing incomplete evidence downstream forces the reader to abstain or guess. Or the evidence may be *too disorganized*: everything needed is present, but it arrives ordered by retrieval score, with the same event extracted several times, planned and actual events interleaved, and occurrence time mixed with mention time, so the reader must count, deduplicate, order, and compare over an unstructured pile, which is where errors concentrate on multi-session and temporal-reasoning questions. HMS closes both gaps on the retrieval side, with a sufficiency verifier for the first and a structured organization layer for the second. Both stages obey the action space of Sec. 3.2: they reallocate or reshape evidence that exists in $\mathcal{M}^*$ and can never re-extract, rewrite, or exceed its coverage.

Sufficiency verification. The planner exposes the answer-relevant slots its plan intends to fill, including the resolved time window, the bridge entities, the fact whose latest value is sought, or the fact whose conflicting values must be surfaced. The verifier $V(C_q, A)$ measures which slots are covered by the candidate set C_q . A slot is deemed covered if the candidate set contains retrieved evidence that satisfies its specific criterion completely. For a time window slot, coverage requires at least one record whose timestamp falls within the resolved interval. For a bridge entity slot, coverage requires a surface match or a co-referent mention of that entity. For a latest-value fact slot, coverage requires the record with the most recent timestamp for the specified key. The verifier then returns an insufficiency signal, namely the set of uncovered slots rather than a free-form judgment. This keeps the decision measurable and tied to the plan rather than to the reader’s eventual confidence. If slots remain uncovered, the verifier triggers a retry confined to the admissible action space. It may enable an additional operator targeting the uncovered slot, relax that operator’s parameters, or raise its budget b_o , then re-fuse. It would never request re-extraction. So when extraction lacks the required evidence, the verifier exits and the uncovered slots are reported. Retries are capped at a small constant number of rounds to bound latency, and we report triggering rate, retry depth, and cost in Sec. 4.

Viewed as a whole, the verifier and its bounded retry constitute a self-correcting retrieval loop: rather than committing to a single recall pass, HMS diagnoses its own coverage gaps against the plan and adapts its retrieval behavior in response, then re-executes. Two properties make this loop safe. It is driven by the measurable insufficiency signal of the coverage check rather than by the reader’s downstream confidence, and it stays within the retrieval-only action space, so it can only reallocate retrieval effort over the frozen bank. We deliberately keep this adaptation on the retrieval side. The verifier decides whether to retrieve more, never whether to rewrite an answer, and it is therefore distinct from answer-side verification, which re-judges a generated answer. We find answer-side verification unstable, as it revises correct answers into over-constrained ones (Sec. 4), and HMS avoids it entirely. The sufficiency verifier carries none of that risk because it cannot alter a generated answer at all.

Gated organization. Once evidence is verified sufficient, the remaining question is its shape. Organization is applied selectively, only where its structuring benefit outweighs its cost, gated on a predicate $g(q, y)$ that is true when two conditions hold jointly,

$$g(q, y) = \mathbf{1}[y \in \mathcal{Y}_{\text{agg}}] \cdot \mathbf{1}[\mathcal{K}(q) \neq \emptyset],$$

where $\mathcal{Y}_{\text{agg}} = \{\text{multi-session, temporal-reasoning, knowledge-update}\}$ collects the question types whose answers require aggregating or reconciling evidence across units, and $\mathcal{K}(q)$ is the set of *reasoning markers* the query matches from a fixed lexicon $\mathcal{K}$ spanning six semantic categories: counting, quantity or amount, absolute or relative time, ordering, state update, and comparison. Organization fires only when $g(q, y) = 1$. Queries failing either condition, in particular simple single-session lookups, bypass it, since unnecessary evidence expansion demonstrably harms them (Sec. 4).

Ledger construction. Let $\mathcal{F}_q$ be the fact list from the verified evidence $\mathcal{E}_q^{\text{raw}}$ and $T(q)$ the non-stopword content terms of the query. For each fact f , we compute a question-overlap count and a reasoning-signal indicator,

$$o(f, q) = \sum_{t \in T(q)} \mathbf{1}[t \in f], \quad h(f) = \mathbf{1}[f \text{ carries a numeric, temporal, monetary, or update signal}].$$

A fact enters the ledger when it either carries such a signal or overlaps the question sufficiently, and is scored so that reasoning-bearing facts dominate,

$$f \in \mathcal{L}_q \iff h(f) = 1 \vee o(f, q) \geq \theta_o, \quad s_{\text{org}}(f, q) = \beta_h h(f) + o(f, q),$$

with a small overlap threshold θ_o and a signal weight $\beta_h > 1$ that prioritizes facts bearing dates, amounts, or ordering cues, those most often needed for hard LongMemEval questions. Because extraction may emit several near-identical facts from one source, we deduplicate conservatively on a source-and-prefix key that pairs the source document with a normalized text prefix, keeping the first occurrence. This removes obvious repeats without merging genuinely distinct events, and it operates only over already retrieved units, so it stays inside the frozen-bank action space.

Each retained row is rendered with explicit metadata,

$$\ell_i = (t_i^{\text{occ}}, t_i^{\text{men}}, \sigma_i, \tau_i, c_i, x_i),$$

namely occurrence time, mention time, source document, fact type, source chunk reference, and compact evidence text. The ledger retains the top rows by s_{org} under a row budget N_{row} , drawn from the top N_{scan} facts through at most N_{cand} intermediate rows.³ Because compression can drop whether an event was planned or completed, or what a number refers to, each ledger row is paired with raw source snippets: over the unique chunks $\mathcal{U}_q = \text{Unique}(\{c_i\})$ we render the first N_{chunk} compact snippets $\mathcal{S}_q$. The reader thus receives two aligned levels,

$$\underbrace{\mathcal{L}_q}_{\text{structured reasoning view}} + \underbrace{\mathcal{S}_q}_{\text{raw source grounding}},$$

the first supporting counting, ordering, and comparison, the second preserving local context for ambiguous cases. Like the verifier, this stage introduces no second generation pass and no answer-side judge. It changes only the shape of the evidence the reader sees, which is why it avoids the instability of answer-side verification (Sec. 4).

Diagnosis-driven self-evolution. The full configuration, **HMS-self-evolve**, adds a controller that specializes the above stages per query at inference time. Keyed to the question type and the reasoning markers already exposed by the plan, the controller escalates recall and snippet budgets, tightens ledger row selection, and applies six deterministic evidence-handling controls, namely exact-date backfill, relative-date grounding, count/total deduplication, current/previous state arbitration, amount/difference calibration, and dual-level source grounding. Each control is distilled from a recurring failure mode. All controls operate on already-retrieved evidence before generation and never inspect or rewrite an answer, so they remain within the retrieval-only action space of Sec. 3.2. Removing the controller yields **HMS-base**.

3.6 Design Rationale and Positioning

The unifying principle of HMS is to move reasoning-critical decisions earlier in the pipeline: plan the recall program from the query, budget and execute only the typed operators it calls for, verify coverage before generation, and organize the verified evidence into a reasoning-ready form, all without ever leaving the retrieval-only action space. Planning grounds underspecified questions and prunes irrelevant channels. Typed operators recover context that is relevant but query dissimilar. The verifier closes the loop on the retrieval side. The organization layer ensures the reader sees evidence it can actually count, order, and compare over.

Relation to prior memory systems. HMS shares its hybrid-retrieval backbone with recent agent-memory systems that fuse semantic, lexical, temporal, and graph signals under a token budget (Latimer et al., 2025; Chhikara et al., 2025). We differ along four axes. First, those systems ground time and entities at ingestion time, whereas HMS grounds the query and selects operators at retrieval time, so one frozen bank can be queried with different recall programs. Second, HMS treats evidence expansion as a library of typed operators with entry-point score propagation and plan-assigned budgets, and makes operator selection an explicit, measurable choice rather than a single generic association channel. Third, HMS closes the retrieval loop with a sufficiency verifier constrained to the frozen bank, turning “is the evidence enough” into a bounded retrieval-only decision rather than a burden on the generator. Fourth, HMS adds a post-retrieval organization layer that reshapes evidence into a structured, source-grounded ledger, which we find more stable than answer-side correction. We include the strongest such prior system as a baseline under identical $\mathcal{M}^*$ and backbone in Sec. 4.

³The packing constants $N_{\text{scan}}, N_{\text{cand}}, N_{\text{row}}, N_{\text{chunk}}$ and the thresholds θ_o, β_h are fixed across all experiments, and their values are given in the supplementary material.

Cost. The additions are not free, and selective, budgeted execution is what keeps them affordable. Planning adds one analyzer call per query. Typed operators enlarge the candidate set and the rerank load only when the plan selects them and only up to b_o . Each verifier retry repeats fusion and reranking within a capped number of rounds. Organization adds a bounded post-processing pass over already retrieved units with no extra model call. Queries with no special structure run a single operator, never retry, and skip organization. We report end-to-end latency, candidate-set size, operator-selection frequency, retry rate, and organization-trigger rate per variant in Sec. 4.

What we expect to verify empirically. Each component targets a specific question category, and we structure the ablation accordingly. Removing operator selection, that is, running all operators always, should raise cost without raising accuracy and hurt queries with misleading channels. Removing the per-operator budget in favor of uniform allocation should raise cost and dilute evidence on queries needing concentrated recall. Removing individual operators should hurt their matched categories: L for multi-session, U for knowledge-update, C for conflicting knowledge-update, and E and G for multi-hop. Removing the retrieval-side verifier should hurt under-recalled queries while reducing latency. Removing the organization layer should hurt multi-session and temporal-reasoning most, where counting and ordering dominate. We further contrast the organization layer against answer-side verifier variants, which we expect to be less stable. The full per-component and per-category ablation is reported in Sec. 4.

4 Experiments

To comprehensively evaluate the effectiveness of our proposed HMS, we compare it against several representative conversational agent memory systems, including AgentMemory (rohitg00, 2026), PlugMem (Yang et al., 2026a), Mastra (Mastra, 2026), and Hindsight (Latimer et al., 2025). This comparison aims to assess the advantages of HMS in long-term memory maintenance, cross-session knowledge integration, and complex reasoning within extended dialogues, while identifying the relative strengths and weaknesses of existing approaches.

4.1 Experimental Setup

We conduct experiments on two complementary benchmarks: LongMemEval (Wu et al., 2025a) and LoCoMo (Maharana et al., 2024). LongMemEval focuses on multi-session interactions, emphasizing knowledge updates, information aggregation across sessions, and temporal reasoning. It is particularly suited for evaluating a system’s ability to maintain consistency and coherence in evolving user conversations. In contrast, LoCoMo targets single long conversations, testing single-hop, multi-hop, temporal, and open-domain reasoning over extensive dialogue histories. Together, these benchmarks capture memory consolidation across sessions and complex reasoning within sessions.

For fair and reproducible comparison, we adopt a unified evaluation pipeline across all systems. Concretely, for each memory system, we (1) ingest the complete conversation history to construct its internal memory store, (2) retrieve relevant memories for each query, and (3) generate the final response based on the retrieved memories. This standardized procedure ensures that performance differences primarily reflect the design of the memory mechanisms rather than variations in evaluation protocols.

Among the baselines, AgentMemory is primarily designed as an indexing and retrieval engine without a built-in conversational interface. To enable consistent evaluation based on dialogues, we developed a lightweight chat wrapper that feeds its retrieved memories into the context along with the user query, instructing the model to answer exclusively based on the provided information.

Finally, we standardize on GPT-5-mini as the backbone model for memory ingestion, response generation, and automatic evaluation. This choice is motivated by two considerations. First, prior studies often employ different models across stages or systems, making cross-method comparisons unreliable due to varying model capabilities. Second, some competing systems leverage more powerful long-context models (e.g., Claude-Opus-4.6), which can partially compensate for suboptimal retrieval by reasoning over noisy or lengthy contexts. By fixing the backbone to GPT-5-mini, we isolate the contributions of the memory systems themselves, enabling a more controlled and interpretable analysis of their intrinsic strengths and limitations.

4.2 Main Results

Table 1 presents the overall and category-level accuracy for HMS compared to other LLM memory systems on LongMemEval. HMS-self-evolve achieves 92.8% overall accuracy, establishing the best result among all compared systems on this benchmark. Compared to the strongest baseline Hindsight (88.8%), HMS-self-evolve achieves a 4.5% relative improvement in overall accuracy, with particularly strong gains on multi-session aggregation (MS: 87.2% vs. 81.2%) and temporal reasoning (TR: 94.7% vs. 88.7%). Its base variant, HMS-base, also reaches a competitive 92.4% overall, consistently surpassing all prior methods.

Table 1 Comparison of representative LLM memory systems on LongMemEval. All systems use GPT-5-mini for memory ingestion, response generation, and automatic judgment. **Overall** accuracy is the micro-average (total correct answers across all questions divided by total number of questions), not the macro-average of the category accuracies.

System	SS-User	SS-Assistant	SS-Preference	Multi Session	Knowledge Update	Temporal Reasoning	Overall
Agentmemory	94.3	87.5	16.7	48.1	61.5	35.3	55.8
Mastra	97.1	98.2	53.3	71.4	87.2	56.4	77.3
Plug Mem	97.1	87.5	76.7	87.2	89.7	56.4	80.2
Hindsight	95.7	98.2	86.7	81.2	89.7	88.7	88.8
HMS-base	97.1	94.6	96.7	86.5	92.3	94.0	92.4
HMS-self-evolve	97.1	96.4	96.7	87.2	91.0	94.7	92.8

Both HMS-base and HMS-self-evolve demonstrate strong performance across all six LongMemEval categories. The largest gain appears on single-session preference recall, where HMS-base and HMS-self-evolve substantially outperform Hindsight (86.7%) and all other baselines. On knowledge-update tracking, HMS-base attains 92.3%, exceeding Hindsight and Plug Mem (both 89.7%). In multi-session aggregation, HMS-self-evolve ties with Plug Mem at 87.2% and achieves a 7.4% relative improvement over Hindsight. For temporal reasoning, HMS-self-evolve reaches 94.7% (HMS-base: 94.0%), leading Hindsight (88.7%) by a clear margin. On single-session user recall, both HMS variants match the top score of 97.1%; on single-session assistant recall, HMS-self-evolve achieves 96.4% and HMS-base 94.6%, slightly behind the 98.2% of Mastra and Hindsight. The one category where HMS methods do not lead is single-session assistant recall, yet they remain highly competitive.

Table 2 Comparison of representative LLM memory systems on LoCoMo. All systems use GPT-5-mini for memory ingestion, response generation, and automatic judgment. **Overall** accuracy is the micro-average (total correct answers across all questions divided by total number of questions), not the macro-average of the category accuracies.

System	Single Hop Reasoning	Multi Hop Reasoning	Temporal Reasoning	Open Domain Reasoning	Overall
Agentmemory	72.7	37.9	56.7	34.4	60.6
Plug Mem	87.2	69.9	25.9	62.5	69.7
Mastra	93.6	83.7	88.2	79.2	89.7
Hindsight	91.1	88.8	84.4	94.6	92.1
HMS-base	91.5	90.8	84.5	95.4	93.1
HMS-self-evolve	92.0	91.5	85.0	95.5	93.5

Table 2 reports results on LoCoMo, which stresses multi-hop and temporal reasoning in long conversations. Both HMS-base and HMS-self-evolve outperform the compared baselines in overall accuracy. HMS-self-evolve attains the highest overall accuracy of 93.5%, outperforming the strongest baseline Hindsight. HMS-base also demonstrates strong results with 93.1% overall accuracy. The proposed systems excel particularly in multi-hop reasoning and open-domain reasoning. HMS-self-evolve achieves 91.5% on multi-hop reasoning and 95.5% on open-domain reasoning. Similarly strong gains are observed in single-hop reasoning (92.0%) and temporal reasoning (85.0%).

These results confirm the effectiveness of the retrieval layer design and self-evolution mechanism in handling complex reasoning tasks that require integrating information across multiple memory entries and temporal contexts. Notably, the overall accuracy is computed as the micro-average across all questions, providing a more balanced view of performance on the full evaluation set.

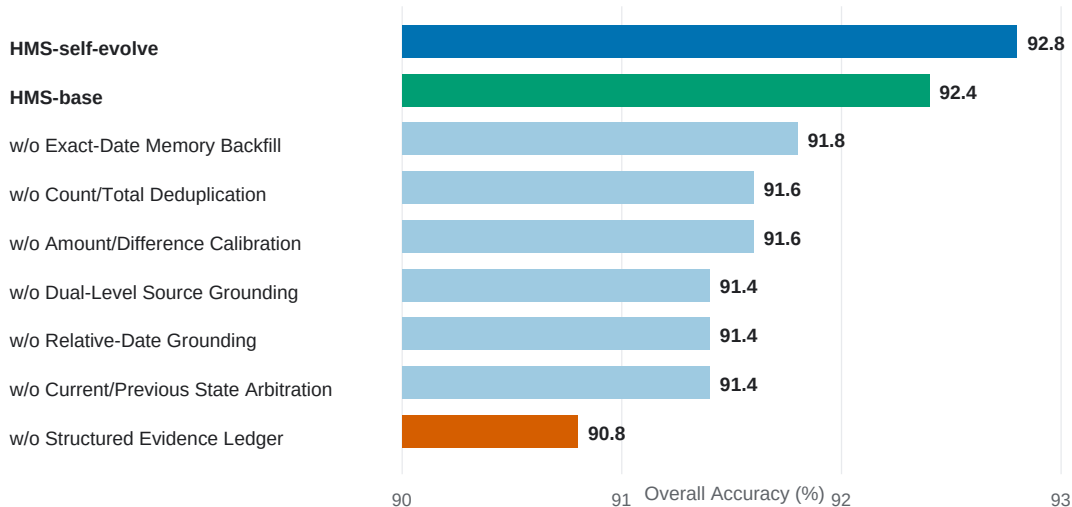


Figure 3 Overall accuracy under ablations of the retrieval-to-reasoning evidence organization layer. HMS-self-evolve achieves the highest overall accuracy among all variants.

4.3 Ablation Study

Ablation Analysis. Figure 3 reports the overall accuracy of HMS-self-evolve and its ablated variants on LongMemEval. HMS-self-evolve achieves the best result at 92.8%, while HMS-base obtains 92.4%. The variants without individual self-evolution controls obtain 91.4–91.8%, and the variant without the Structured Evidence Ledger obtains 90.8%. This ordering shows that the Structured Evidence Ledger provides the main foundation for the system, and the self-evolution controls further refine how the evidence is used during answer generation.

We ablate the components of the retrieval-to-reasoning layer and report overall accuracy on LongMemEval (Figure 3). The question is whether the gains come from the organization mechanisms of Section 3 or could be reproduced by a single prompt instruction.

Removing the Structured Evidence Ledger costs the most, lowering overall accuracy to 90.8%. The ledger exposes temporal cues, numerical values, source documents, and fact types in a compact reasoning view. Without it the answer model receives a flat retrieved context and must recover event structure, duplicate mentions, temporal order, and state changes on its own, which destabilizes long-horizon reasoning even when the relevant memories are present. Removing the Diagnosis-Driven Self-Evolution Controller yields the HMS-base setting at 92.4%, against 92.8% for the full system. The gap is moderate but consistent, as expected for a component that adjusts evidence use downstream of the main organization layer rather than replacing it.

The finer ablations locate where self-evolution helps. Count/Total Deduplication addresses cross-session aggregation, where several extracted facts can describe one event and planned or recommended actions should not be counted as completed ones. Relative-Date Grounding and Exact-Date Memory Backfill address temporal lookups, where expressions such as “last Saturday” or “ten days ago” must be resolved against the question date before evidence is selected. Amount/Difference Calibration permits a numerical comparison only when both sides are explicitly supported, and Current/Previous State Arbitration separates the latest state from superseded ones on update questions. Dual-Level Source Grounding keeps raw snippets next to ledger rows, which matters when a compact fact omits whether an event was actual, planned, or merely recommended.

Case Study. Figure 4 makes two of these effects concrete. In the state-update example, Hindsight defaults to the latest personal-best time, whereas HMS-base and HMS-self-evolve recover the previous state explicitly requested by the query. In the missing-evidence example, Hindsight and HMS-base collapse an unobserved count to 0, while HMS-self-evolve returns Insufficient Information rather than producing an unsupported numerical answer. These cases illustrate that the self-evolution controls affect not only aggregate accuracy but also how the system resolves the queried state and responds when the available memory does not support a definitive answer.

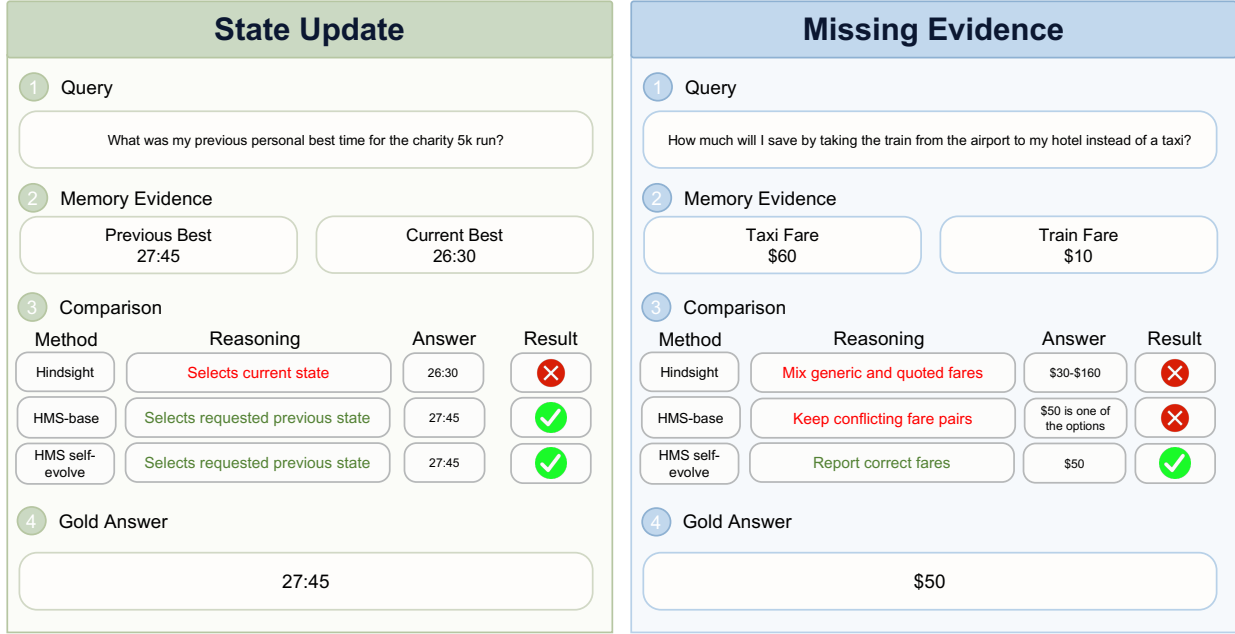


Figure 4 Qualitative comparison of state resolution and evidence sufficiency. The left example shows that HMS selects the historical state requested by the query rather than the latest state. The right example shows that HMS-self-evolve reports insufficient information rather than assigning zero when relevant evidence is missing.

Together, the quantitative ablations and qualitative cases show that HMS benefits from the interaction of scoped retrieval planning, ledger organization, source grounding, and diagnosis-driven control. Among all the tested variants, removing the ledger produces the largest observed drop.

5 Conclusion

We asked how much accuracy retrieval alone leaves on the table in long-horizon conversational memory, and answered it by decoupling retention from recall. Over a fixed memory bank, HMS plans an executable recall program per query, expands evidence through six typed operators, verifies coverage with bounded retrieval-side retries, and organizes the result into a structured ledger. This improves overall accuracy by 4.0% on LongMemEval and 1.4% on LoCoMo under identical stored memories, and ablations attribute each gain to its matched question category. Two findings generalize: relation-typed, budgeted expansion turns the choice of relatedness into a measurable decision, and self-correction is far more stable when it adapts retrieval than when it rewrites answers.

The limits point to future work. Recall cannot exceed the extraction ceiling we report, motivating joint optimization of retention and recall. The planner and verifier rely on question types and hand-specified slots that could instead be learned from retrieval feedback. And the organization schema, built for counting and ordering, invites extension to richer reasoning and to agentic settings beyond conversation.

6 Future Roadmap

The next stage of the Holographic Memory System (HMS) should extend the decisions it currently makes at recall time both to forms of memory that cannot be reduced to stored text and to the operating conditions of deployed consumer systems. The present system keeps retained content fixed, plans retrieval for each query, checks whether the retrieved evidence is sufficient, and organizes that evidence before generation. These mechanisms are a starting point for future work, not evidence that the current implementation already supports world modeling, multiple media, physical control, continuous acquisition, proactive recall, or cross-user governance. In world models, memory must carry persistent hidden state. Across multiple media, it must preserve and connect evidence in text, images, audio, and video. In physical

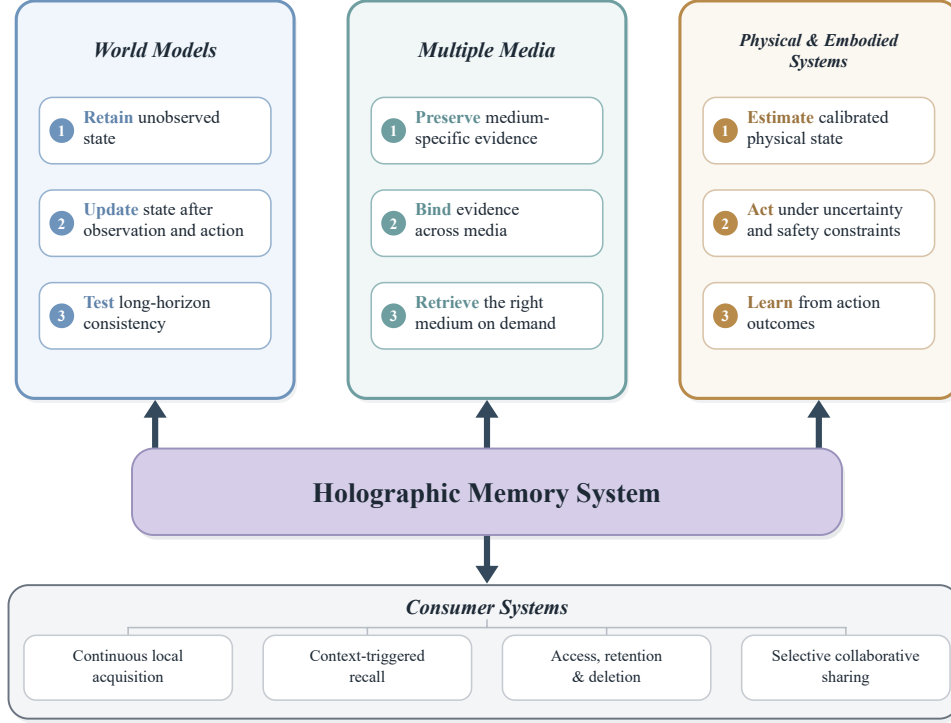


Figure 5 Three research directions for extending HMS beyond static conversational records, together with a cross-cutting consumer deployment layer. World-model memory retains and updates hidden state; cross-media memory preserves and connects evidence in text, images, audio, and video; physical and embodied memory links calibrated state estimates to action and feedback. Consumer systems additionally require continuous local acquisition, context-triggered recall, access, retention, and deletion controls, and selective collaborative sharing.

and embodied systems, it must support calibrated state estimation, action, and correction from observed outcomes. Consumer systems add a cross-cutting deployment layer in which memory is acquired continuously, invoked from context under local resource and privacy constraints, governed by user-visible policies, and shared selectively. Figure 5 summarizes these three capability directions and the consumer layer that spans them.

6.1 Memory as Hidden State in World Models

A world model predicts how an environment will change, including parts of the environment that are absent from the current observation (Ha and Schmidhuber, 2018; Wang et al., 2026a). Its memory should therefore be a persistent hidden state: a learned representation that carries information about the environment across observations and actions. If a camera turns away from an object, the state should preserve the object’s identity and last observed condition while increasing uncertainty about changes it cannot see. If an action changes the object, the state should record the change rather than replay the earlier view. This is different from extending a video context window. More frames can provide additional evidence, but they do not by themselves identify which unobserved facts remain valid, which have changed, or how uncertainty should grow while the environment is not observed. A useful hidden state must therefore encode both an estimate and its uncertainty, so that absence is not mistaken for evidence that the world has stayed unchanged.

Future HMS research should make this hidden state the primary memory for world models. Explicit object records, spatial maps, or selected frames can provide inspectable views of that state, but they should not become a separate memory system that competes with the predictive state. A query or planning objective should read only the state needed for the next prediction or action, while new observations update the same state. Work on scene-consistent context, spatial memory, and joint-embedding prediction provides useful components for this interface (Yu et al., 2025; Wu et al., 2025b; Assran et al., 2025). This query-specific state view should expose the object, region, relation, or uncertainty relevant to the current decision without decoding the entire state back into pixels. The unresolved problem is the update rule: when

observations conflict, the model must distinguish a real environmental change from noise, revise the affected state, and reduce confidence in details that are no longer supported by observations.

The central evaluation should be long-horizon state consistency rather than frame-level plausibility. A model can be tested by leaving a scene, observing other events, and then returning after an object has remained still, moved, or become uncertain. The hidden state succeeds if it supports the correct prediction and action after the absence, not merely if the generated frame looks realistic. The same test can measure whether errors accumulate across action-conditioned rollouts and whether the query-specific state view exposes the state used for a prediction. Existing memory-focused world-model benchmarks provide a starting point for such tests (Zhang et al., 2026b). An HMS extension would add explicit checks for unsupported state, conflicting observations, and correction after action, so that verification operates on the model’s evolving state rather than only on retrieved records.

6.2 Memory Across Multiple Media

Memory across multiple media must preserve what each medium contributes. Text records names, claims, and explicit relations. Images (Han et al., 2026b,a) preserve appearance, layout, and spatial detail. Audio preserves speakers, tone, and environmental sound. Video preserves change and temporal order. A system loses information when it converts all four into a single textual summary, but it also fails when it stores them separately and cannot recognize that they describe the same event or entity. The useful unit is a cross-media episode: a time-bounded record that links an event and its entities to the supporting evidence in each available medium. An episode need not contain every medium, but it must state which media are present and which claims each one supports. This definition keeps the media visible instead of hiding them behind a generic multimodal embedding (Hu et al., 2025b; Long et al., 2025; Liu et al., 2025).

The write path should retain a concise semantic description together with pointers to the relevant text span, image region, audio interval, or video segment. Compact learned features may support retrieval, but the original evidence must remain available when a later question depends on color, position, voice, sound, or motion. At query time, the system should select the medium required by the current query rather than append every stored item to the context. A spoken description may retrieve an image; a textual query may require an audio interval; a visual cue may reopen the video segment in which an event changed. Storage cost, latency, and privacy will differ by medium, so the write policy should decide what remains local, what is compressed, and what can be deleted without losing the evidence needed for later use. Long video is an important stress test because it forces aggressive compression and online writing, but it is one case within the broader cross-media problem rather than the organizing principle for the section (Song et al., 2024; Sun et al., 2025).

Cross-media memory must also change when later evidence corrects an earlier interpretation. Each stored claim should identify which source supports it, which parts were inferred, and which later observation revised it. Evaluation should therefore go beyond locating a relevant segment. A test should ask a question in one medium, require evidence from another, and verify that the returned native evidence supports the answer. It should also introduce conflicting or incomplete evidence and check whether the episode is updated without erasing valid history. Benchmarks for multimodal conversational memory already expose parts of this problem (Bei et al., 2026). A future HMS implementation should measure cross-media retrieval accuracy, evidence fidelity, correction after contradiction, and the cost of retrieving the selected evidence. These measures should be reported separately so that compressing all media into a cheap but lossy summary cannot appear equivalent to preserving evidence that the downstream task can inspect.

6.3 Memory for Physical and Embodied Systems

Physical memory begins before storage, when sensor measurements are converted into an estimate of the current world. That estimate is useful only if it records when and where the data was captured, which device produced it, which calibration was in effect, how uncertain it is, and how long it can be reused. Sensor bias, clock mismatch, mounting changes, and wear can make an old estimate unsafe even when it remains semantically relevant (Lee, 2008; Hurst et al., 2026). Smart glasses and wearable assistants make this problem concrete: camera alignment, inertial measurements, audio, and user context may all drift over extended use. When sensors disagree, the system should preserve the conflict or withhold the estimate rather than silently convert an unstable signal into durable state. The system should lower its confidence, request recalibration, or refuse reuse when those conditions no longer support the estimate, while keeping sensitive raw streams on the device when possible (Yang et al., 2026b).

The next step is to use the physical-state estimate in an action under explicit time, uncertainty, and safety constraints. Industrial robots and digital twins need state that remains aligned across sensors, controllers, simulations, and

operators (Zhou et al., 2026). Autonomous vehicles face the same requirement under stricter deadlines: a relevant obstacle estimate is unusable if it arrives late or no longer matches the current scene. Vision-language-action (VLA) policies and world action models provide complementary interfaces for this loop. A VLA policy proposes an action from perception and instruction, while a world action model predicts the state that an action should produce (Brohan et al., 2023; Wang et al., 2026b). Memory should supply the current estimate and relevant interaction history without replaying entire trajectories. It must also separate task-level experience from assumptions about body geometry, payload, tool, contact conditions, and operating environment. An action that succeeded for one robot or site should not become a transferable skill until the system can state which of those conditions still hold.

Action closes the loop by producing evidence that can confirm or correct memory. The system should compare the predicted and observed outcomes, update the physical-state estimate, and retain the part of the interaction that explains success, failure, or recovery. A failed grasp, unexpected slip, or human intervention can be more informative than another successful trajectory because it identifies where an earlier estimate or action assumption was wrong. Memory-augmented VLA research shows why this history matters when task-relevant evidence leaves the current view (Shi et al., 2025). The update path should distinguish immediate state correction from slower updates to reusable skills, simulation data, or model training; a long-lived agent cannot retrain after every episode, but it should not discard repeated evidence of the same failure. The research target is a complete sense-state-act-update cycle, not a larger sensor archive. Evaluation should measure calibration under drift, action quality under uncertainty, recovery after prediction error, and whether each update remains traceable to physical evidence. Lifecycle and timing requirements established for cyber-physical systems provide a systems-level basis for those tests (NIST CPS Public Working Group, 2017).

6.4 Memory for Consumer Systems

Conversational memory benchmarks begin with a retained bank and an explicit question, whereas a consumer system must build and revise memory continuously from calendars, messages, browser activity, documents, and application events (Wu et al., 2025a; Maharana et al., 2024). This changes the write path before it changes retrieval. Structured sources can be handled by deterministic adapters, and unstructured material can be converted into typed memory units by a small local model. Raw source data should remain on the device by default. Only when the user explicitly enables remote processing should the system send task-bounded summaries or features for heavier cross-source reasoning. Work on bounded-compute memory shows how retrieval, writing, and consolidation can be divided between lightweight online processing and heavier offline maintenance (Zhang et al., 2026a), while executable memory operations provide a concrete basis for incremental addition, revision, and deletion (Zhao et al., 2026). This split between on-device work and explicitly enabled remote work also has a deployment precedent outside memory systems (Apple Security Engineering and Architecture (SEAR) et al., 2024). Each memory unit should record its source, capture time, confidence, owner, retention conditions, and derivation history, extending the current HMS schema rather than assuming that heterogeneous consumer data already fits it unchanged.

Consumer recall also begins before the user asks a question. A change in the active document, an approaching calendar event, a participant joining a conversation, or a return to an interrupted task can be encoded as a candidate pseudo-query. A low-cost trigger gate should first decide whether the context warrants planning at all. When it does, the existing recall planner, typed operators, and sufficiency verifier can remain the recall core: temporal proximity can parameterize the Temporal operator, a recognized participant can seed Entity-Bridge, and topic discontinuity can invoke Session-Local context. A separate delivery gate should then decide whether the recalled evidence should be surfaced now, prepared silently, or withheld. This separation prevents every context change from becoming either an expensive retrieval or an interruption. Proactive-assistance research similarly treats timing as a decision that balances expected usefulness against interruption cost (Pu et al., 2025). Accepted suggestions and dismissals can adjust future trigger thresholds without turning the user’s attention policy into another stored fact about the world.

Persistent personal memory requires governance at the same granularity as its memory units. Each memory unit should carry owner, visibility, retention, provenance, and revocation metadata, and those constraints should be applied before candidate generation and graph traversal so that restricted evidence cannot influence neighboring scores. Deletion or revocation should invalidate direct entries and propagate through retrieval indices, derived summaries, and authorized copies; this is a systems requirement for future HMS work, not a guarantee of the current retrieval-only implementation. Collaboration involves multiple users with separate permissions. Users should share selected memory units rather than entire banks, preserve origin attribution, and surface conflicting accounts jointly instead of silently preferring one person’s version. Multi-party memory benchmarks already expose the need to track roles, evolving decisions,

and cross-participant attribution (Hu et al., 2026b); selective shared memory with role-based access control has also improved completion in a specific enterprise code-generation setting (Pedada et al., 2026). Under this extension, the Contradiction operator can expose authorized cross-user disagreements and the verifier can expand into a teammate’s shared scope only when the plan and access policy both permit it. The HMS recall core therefore remains intact, but it is surrounded by new acquisition, triggering, policy-enforcement, and collaborative-sharing components.

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Appendix

A Contributors

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B Benchmarks and Compared Systems

B.1 Benchmarks

LongMemEval (Wu *et al.*, 2025a). LongMemEval evaluates chat assistants on long-term interactive memory over multi-session user histories. Each question is paired with a query date and falls into one of six categories: single-session user recall, single-session assistant recall, single-session preference, multi-session aggregation, knowledge update, and temporal reasoning. The benchmark stresses whether a system can resolve relative time expressions, aggregate evidence scattered across sessions, and track facts that change over time.

LoCoMo Maharana *et al.* (2024). LoCoMo targets very long single conversations between two speakers and tests memory within an extended dialogue rather than across separate sessions. Its questions span single-hop, multi-hop, temporal, and open-domain reasoning over the conversation history. The benchmark emphasizes compositional reasoning over many dialogue turns under a fixed context budget.

B.2 Compared Systems

AgentMemory (rohitg00, 2026). AgentMemory is an open-source indexing and retrieval engine for agent memories that stores extracted records and serves them through standard similarity search. It does not ship a built-in conversational interface, so we evaluate it through a lightweight chat wrapper that feeds its retrieved memories into the context together with the user query (Sec. 4.1).

PlugMem (Yang *et al.*, 2026a). PlugMem is a task-agnostic plugin memory module designed to attach to an existing LLM agent without modifying the agent itself. It maintains a persistent record store and exposes retrieval over that store as a drop-in component, emphasizing portability across tasks and backbones.

Mastra (Mastra, 2026). Mastra’s observational memory maintains a continuously updated, compressed observation log of the conversation and answers queries by reading over this curated summary state rather than performing per-query evidence retrieval. It represents the summary-centric point in the design space, in contrast to record-level retrieval systems.

Hindsight (Latimer *et al.*, 2025). Hindsight is a retain–recall–reflect memory system that distills each session into stand-alone memory units, wires them into a graph with entity, temporal, semantic, and LLM-asserted causal edges at ingestion time, and recalls over this graph with a single spreading-activation pass. It is the strongest prior system in our comparison, and its retention stage is what HMS adopts unchanged to build the frozen memory bank (Sec. 3.1).

C Complete Case Showcase: Process Traces and Error Localization

This appendix walks through eight LoCoMo questions on which HMS answers correctly and Hindsight does not, tracing each case end to end: the question and its required answer elements, the ground-truth evidence chain in the original dialogue, what each system retrieved (with ranks in its own candidate list), what it answered, and the pipeline stage at which the failure occurred. Cases come from the evaluation run of Sec. 4 (385 questions). Correctness follows an element-level manual audit: a case is retained only if HMS covers every required gold element and Hindsight misses

Table A Overview of showcase cases. “Locus” names the pipeline stage at which Hindsight’s error occurs, determined from its retrieval log and answer; “Component” names the HMS mechanism that repairs it. B/SE marks whether HMS-base / HMS-self-evolve is correct.

#	Question (abbrev.)	Failure locus	HMS component	B/SE
1	Martial arts John has done	retrieval miss (cross-session)	entity-bridge expansion	✓/✓
2	U.S. regions visited/planned	retrieval miss + channel noise	expansion + plan pruning	✓/✓
3	European countries visited	retrieval miss (cross-session)	entity-bridge expansion	✓/✓
4	Family activities on hikes	retrieved but unused in answer	structured ledger	✓/✓
5	How long ago: 18th birthday	wrong temporal anchor	planner time resolution	✓/✓
6	Dinner-spread dishes	atomic fact too coarse	dual-level source grounding	✗/✓
7	Plan to honor pet’s memory	wrong event selected as answer	state/intent arbitration	✗/✓
8	Plan for evening of Jul 7	same-session late turn missed	session-local + time window	✗/✓

at least one; benchmark judge labels are reported but do not override the audit. No qualifying case existed for the temporal-reasoning category in this run, and we do not fabricate one. Where only HMS-self-evolve is correct we show all three systems, since these cases isolate the self-evolution controls (Sec. 4.3).

Table A localizes each failure. The pattern across cases mirrors the failure taxonomy of Sec. 1: evidence never retrieved (Cases 1–3), evidence retrieved but unused (Case 4), relative time resolved against the wrong anchor (Case 5), atomic facts too coarse for the requested detail (Case 6), and evidence present but the wrong event selected as the answer (Cases 7–8).

C.1 Evidence Never Retrieved (Cases 1–3)

Case 1 Cross-session enumeration: two martial arts, one retrieved

Question. What martial arts has John done? **Gold.** Kickboxing, Taekwondo.

Ground-truth evidence chain.

[session_1, 2022-12-17, D1:4] John: “I’m doing kickboxing and it’s giving me so much energy.”
[session_2, 2022-12-22, D2:28] John: “. . . I’m off to do some taekwondo!”

Process. The two facts sit five days apart in different sessions and share no surface tokens with each other. Hindsight’s candidate list contains only the kickboxing units (ranks 48 and 81 of 105); the taekwondo unit never enters its context. HMS retrieves both: kickboxing at ranks 5 and 8, and the taekwondo unit at rank 33, reached by expanding from the strong John/kickboxing entry points along the shared entity rather than by query similarity.

Hindsight. ✗ “Kickboxing.” — and its reasoning asserts “no other martial art (e.g., karate, taekwondo, MMA) is mentioned in the provided memories,” a confident claim made over an incomplete retrieved set.

HMS. ✓ “Kickboxing (practiced). John also said he was going to do taekwondo (stated on 2022-12-22), but the memories don’t explicitly confirm he completed taekwondo.”

Where the error is. Pure retrieval miss: the answer-bearing unit is relevant but dissimilar to the query, and top- k similarity never surfaces it. HMS additionally keeps the planned/completed distinction explicit via ledger fact typing instead of flattening it.

Case 2 Cross-session union: places visited and places planned

Question. What areas of the U.S. has John been to or is planning to go to? **Gold.** Pacific Northwest, East Coast.

[session_11, 2023-04-10, J] John: “We explored the coast up in the Pacific Northwest and hit some cool national parks.”
[session_12, 2023-04-18, J] John: “I’m planning a trip to the East Coast.”

Process. The question asks for the union of a completed trip and a planned one, recorded in adjacent sessions. Hindsight retrieves the East Coast plan (rank 20) but not the Pacific Northwest unit, and instead pads the answer with

topically associated but non-responsive material (West County flooding, a hometown renovation). HMS retrieves both required units (East Coast rank 22, Pacific Northwest rank 28) and reports the union with the been-to / planning distinction intact.

Hindsight. ✗ “East Coast — planning a trip. . . West County (his old area). . . Hometown — he helped renovate a community center. . .” (misses Pacific Northwest; adds unrequested regions).

HMS. ✓ “Been to: the Pacific Northwest coast (road trip in 2022, visited national parks). . . Planning/intent: a trip to the East Coast (planning mentioned 2023-04-18). . .”

Where the error is. A retrieval miss compounded by noise: a required unit is absent while confidently associated but irrelevant units fill the budget — precisely the “wasteful and sometimes harmful” channel behavior that per-query operator selection is designed to prevent.

Case 3 Two trips, two years apart: Spain retrieved, England lost

Question. What European countries has Maria been to? **Gold.** Spain, England.

[session_8, 2023-03-06, J Maria: “I got the idea from that trip to England a few years ago — I was mesmerized by the castles.” / “It was in London.”

[session_13, 2023-05-04, J Maria: “Last year I took a solo trip and took this pic in Spain.”

Process. The England evidence is doubly indirect: it appears as a reminiscence inside a conversation about home decoration, and the country is named only through London. Hindsight’s list contains a single relevant hit (Spain, rank 89); no England/London unit is retrieved. HMS recovers both (Spain rank 27, London rank 99), the London unit entering through entity expansion from Maria rather than through similarity to the query.

Hindsight. ✗ “Spain (solo trip in 2022).”

HMS. ✓ “Spain (solo trip in 2022) and the United Kingdom — specifically London (mentioned in conversation on 2023-03-06).”

Where the error is. Retrieval miss on evidence whose topical frame (interior decoration) differs entirely from the question’s frame (travel); only relation-typed expansion, not similarity, reaches it.

C.2 Evidence Retrieved but Unused (Case 4)

Case 4 The decisive details sit at ranks 6, 15, and 24 — and never reach the answer

Question. What does Melanie do with her family on hikes? **Gold.** Roast marshmallows, tell stories.

[session_10, 2023-07-20, J Melanie: “We roast marshmallows, tell stories around the campfire and just enjoy each other’s company.”

[session_16, 2023-09-13, J Melanie: “We roasted marshmall[ows]. . .”

Process. This case is the mirror image of Cases 1–3. Hindsight’s own candidate list *contains* the marshmallow and story units at ranks 6, 15, 24, and 44 — retrieval succeeded. But the units reach the reader as part of a flat, score-ordered pile dominated by generic hiking/nature/bonding material, and the generated answer reports only that generic layer. HMS retrieves the same units (ranks 3, 13, 29, 44) and, because the aggregation-style query fires the organization gate, the ledger’s signal-weighted row selection promotes the concrete activity facts into the reasoning view the reader sees.

Hindsight. ✗ “She hikes in the mountains and forests with her family to explore nature, hike to scenic viewpoints, relax and bond—creating family memories. . .” (both required elements missing despite being retrieved).

HMS. ✓ “. . . on these outings they bond (telling stories), sit around campfires and roast marshmallows. . .”

Where the error is. Not retrieval: an evidence-*utilization* failure in the answer stage, induced by evidence shape. It is direct support for the paper’s claim that retrieval alone is insufficient when the recalled set reaches the reader disorganized.

C.3 Wrong Temporal Anchor (Case 5)

Case 5 “Ten years ago” in the memory, “13 years ago” in the answer

Question. How long ago was Caroline’s 18th birthday? **Gold.** 10 years ago.

[session_4, 2023-06-27, J Caroline: “. . . my hand-painted bowl. A friend made it for my 18th birthday ten years ago.”

Process. Both systems retrieve the same single evidence unit (Hindsight rank 17; HMS rank 31), which anchors the birthday to 2013 and states the elapsed time relative to the 2023 conversation. The divergence is purely in temporal grounding: Hindsight recomputes the interval against the evaluation machine’s wall-clock date (2026-07-14), yielding 13 years; HMS resolves the question against the conversational anchor, where occurrence time (2013) and mention time (2023-06-27) are kept as separate ledger columns, and reports the 10-year interval the dialogue itself asserts.

Hindsight. ✗ “About 13 years ago. . . as of 2026-07-14 it was roughly 13 years ago.”

HMS. ✓ “Her 18th birthday was in 2013 — about 10 years before the memory mention on 2023-06-27.”

Where the error is. Anchor confusion: run time silently substitutes for question time. Identical evidence, opposite outcomes — the failure is entirely in how time is grounded, the recall planner’s first responsibility.

C.4 Atomic Fact Too Coarse (Case 6)

Case 6 The extracted fact says “made dinner”; the menu lives one turn later in the source

Question. What kind of food did Maria have on her dinner spread with her mother? **Gold.** Salads, sandwiches, homemade desserts.

[session_13, 2023-05-04, D13:16] Maria: “My mom and I made some dinner together last night!”

[session_13, 2023-05-04, D13:18] Maria: “It had lots of great things like salads, sandwiches, and homemade desserts.”

Process. Extraction distilled the event into an atomic fact — Maria and her mom made dinner on May 3 — without the menu, which appears in the immediately following turn of the raw dialogue. Systems that see only the fact layer therefore conclude, reasonably but wrongly, that the information does not exist. HMS-self-evolve pairs ledger rows with raw source snippets (dual-level grounding), and the expanded source context of the dinner fact contains the menu turn verbatim.

Hindsight. ✗ “The memories do not specify what was on the dinner spread.”

HMS-base. ✗ “The memories do not specify the food.”

HMS-self-evolve. ✓ “Her dinner spread. . . included salads, sandwiches, and homemade desserts.”

Where the error is. Neither retrieval nor generation in the usual sense: the fact layer is genuinely silent, and only a system that can drop to source grounding recovers the detail. The three-way contrast isolates dual-level source grounding as the deciding control, consistent with its ablation row (Sec. 4.3).

C.5 Evidence Present, Wrong Event Selected (Cases 7–8)

Case 7 The rescue-dog unit is in everyone’s top 25 — only one system answers with it

Question. How does John plan to honor the memories of his beloved pet? **Gold.** By considering adopting a rescue dog.

[session_17, 2023-06-03, J John shares a cherished photo of Max (current memorial act).

[session_17, 2023-06-03, J Maria: “Any plans to give another pet a loving home?”

[session_17, 2023-06-03, J John: “We’re considering adopting a rescue dog — for love and to teach our kids responsibility and compassion.”

Process. All three systems retrieve the rescue-dog unit (rank 22–23 in every candidate list), so retrieval is not

the discriminator. The question asks for a *future plan*; the session also contains a salient *current* memorial act (sharing Max’s photo). Hindsight and HMS-base both answer with the photo, i.e., they select the wrong member of a set of same-topic units in different states. HMS-self-evolve’s state arbitration separates the current act from the prospective intention and reports the plan the question asked for.

Hindsight. ✗ “He plans to honor Max by sharing and keeping cherished photos/memories of him. . .”

HMS-base. ✗ “He honored Max by sharing a cherished photo of Max. . .”

HMS-self-evolve. ✓ “. . . He and his family are also considering adopting a rescue dog to give it a loving home and to teach their kids r[esponsibility]. . .”

Where the error is. An answer-selection failure over correctly retrieved evidence: without an explicit current-vs-prospective distinction, the most salient unit wins over the responsive one. The base/self-evolve split again localizes the repairing control.

Case 8 Same day, later turn: the plan is the last thing Maria says

Question. What did Maria plan to do later on the evening of 7 July, 2023? **Gold.** Have dinner with friends from the gym.

[session_23, 2023-07-07 18:29, J Maria: “Well, I’m off to have dinner with some friends from the gym. Talk to you later!”

Process. The question pins a date and asks for the *later* plan. That evening’s session is dominated by an earlier topic — a community meeting about flood recovery — and the target plan is Maria’s sign-off in the session’s final turn. Hindsight anchors on the dominant topic, finds no “plan” there, and reports that no plan is recorded; HMS-base similarly answers with the flood meeting. HMS-self-evolve’s session-local expansion within the resolved date window reaches the late-turn evidence and answers with it.

Hindsight. ✗ “The memories do not record any explicit plan by Maria for later on the evening of 7 July 2023. . .”

HMS-base. ✗ “Attend the meeting John was organizing that evening. . .”

HMS-self-evolve. ✓ “She planned to have dinner with some friends from the gym that evening (July 7, 2023).”

Where the error is. A same-session localization failure: the responsive unit is temporally in-window and session-adjacent to strong entry points, but topically overshadowed. Position-keyed expansion, not similarity, recovers it — and the abstention (“no plan recorded”) shows how silently this failure mode passes to the reader.

Summary. Across the eight cases, the error never lies in the stored memory: every gold element is recoverable from the frozen bank, as the ground-truth chains show. What differs is where each system loses it — before retrieval (temporal anchoring, Case 5), during retrieval (Cases 1–3), after retrieval in evidence shape (Case 4), below the fact layer (Case 6), or at answer selection over correctly retrieved units (Cases 7–8). Cases 6–8, where HMS-base fails alongside Hindsight and only HMS-self-evolve succeeds, provide the qualitative counterpart to the self-evolution ablation of Sec. 4.3: each is repaired by exactly one named control (dual-level source grounding, state arbitration, session-local recovery), and the repair operates on evidence handling rather than on rewriting answers.